

Future History: Tech Levels and Design Framework

Sources: *GURPS Ultra-Tech* (pp. 5–17), *GURPS Future History* (pp. 4–37), *2300AD Core Rulebook* (pp. 2–76), *GURPS Transhuman Space Classic* (pp. 26–220)

1. The TL Framework

Tech Level (TL) is a number indicating a society's technological sophistication. It is a shorthand for the entire suite of materials, manufacturing, medicine, energy, transport, and weapons available at a given time. Within any setting, different regions, factions, or institutions may sit at different TLs simultaneously.

GURPS TL scale (relevant range):

TL	Name	Paradigm technology	Approx. Earth start (medium pace)
8	Digital Age	Computers, nuclear power, space probes	~1980
9	Microtech Age	Nano-composites, wearables, quantum computing, neural interfaces	~2030
10	Robotic Age	Volitional AI, molecular nanotechnology, fusion reactors for spacecraft	~2120
11	Exotic Matter Age	Mature nanotechnology, sub-atomic engineering, antimatter propulsion	~2500

TL	Name	Paradigm technology	Approx. Earth start (medium pace)
12	Miracles Age	Full nano/exotic matter integration, self-replicating swarms, living metal	~4000

Progression rates vary. A slow pace doubles or triples these dates; accelerated halves them. Superscience (FTL, gravity control, etc.) can appear at any TL as a discontinuous breakthrough.

TL Progression Start Dates (from TL8 baseline ~1980):

TL	Accelerated	Fast	Medium	Slow	Retarded
9	2020	2025	2030	2040	2050
10	2050	2075	2120	2200	2500
11	2100	2200	2500	3000	7000
12	2200	2600	4000	8000	20000

These dates are Earth-centric. Colony worlds typically lag the metropole by 1–3 TLs; isolated settlements may lag further.

2. TL Descriptions

TL9 — Microtech Age

Key capabilities: Nano-scale composite materials, polymer electronics, printed computers, wearable computing, chameleon suits, quantum computers, neural interfaces, mobile robots, androids (expensive), antimatter manufactured routinely (small quantities), powered exoskeletons and battlesuits, tissue engineering and organ growth, extended lifespan.

Character of the age: Human-scale biology still dominates but begins to be augmented. Robots exist but are expensive; AI is task-specific rather than general. Energy comes from fuel cells, advanced gas turbines, and early fusion. Information density explodes. Governments still recognisable as nation-state descendants. Population expansion slows as urban density and biotech reduce birth rates.

Superscience possibilities: FTL drives (parachronic reactionless), psi-augmentation devices.

2300AD analogues: Older surplus weapons (SG-77 assault rifle, P-11 pistol, Wolf shotgun, MG-7 machinegun) are TL9 items still in widespread use on the frontier.

TL10 — Robotic Age

Key capabilities: Volitional AI — first truly thinking machines. Inexpensive sapient machines. Micro-robot swarms. Biomechanical nanomachines for medicine. Molecular nanotechnology in manufacturing (assembly from bottom up, not just top-down machining). Pseudo-alive self-repairing polymers ("living plastics"). Power cells sufficient for electromagnetic weapons and high-energy lasers. Fusion reactors small enough for spacecraft.

Character of the age: The volitional AI singularity risk is present but not inevitable. Mass automation displaces most human labour in physical manufacturing; human value lies in creativity, social trust, and novel problem-solving. Cities restructure around robot logistics. Interplanetary travel is practical and commercial. Simple terraforming (Mars-class) becomes possible in principle. Political units remain planetary or sub-planetary.

Superscience possibilities: Gravity control, contragrav vehicles, reactionless drives.

Energy throughput: Approximately 128× human metabolic rate per person (same as TL7–8 but applied more efficiently).

Population dynamics: Expansion slows (~0.7%/decade) as habitat-building absorbs capital that would otherwise go to reproduction.

TL11 — Exotic Matter Age

Key capabilities: Precise atomic structure control. Mature molecular nanotechnology. Nanofactories producing diamondoid components. Sub-atomic engineering with exotic materials (muon substitution for hyperdense matter). Antimatter as compact energy storage *and* rocket propulsion (practical interstellar). Portable particle beams and X-ray lasers. Diamondoid composite armour. Nanotechnology dominates medicine — chrysalis machines reassemble bodies at the cellular level. Mind copying without destroying the original.

Character of the age: Human biology becomes a design space rather than a constraint. Radical life extension practical; identity and succession become political flashpoints. Interstellar travel possible (generation ships or near-light antimatter drives; FTL still superscience). Industrial nanofactories begin to undercut conventional mining and manufacturing. Social stratification shifts from wealth to access to nanofactory capacity.

Superscience possibilities: Contragrav miniaturised for personal vehicles, force screens, hyperdense armour, compact nuclear reactors, matter transmission (sender and receiver required), neural disruptors, recorded minds.

Population: Growth resumes (~3.5%/decade) as life extension offsets declining birth rates.

TL12 — Miracles Age

Key capabilities: Full nanotechnology and exotic matter integration. Living metal. Gamma-ray lasers. Self-replicating nanomachine swarms. Body regeneration in hours. Entire planets can be moved or disassembled. Most civilisation run by sentient machines. Raw materials drawn from asteroid belts and gas giants as routine industry.

Character of the age: Scarcity of physical goods effectively ends for any society that reaches this point. The binding constraints become energy, time, and social coordination. Planetary engineering is practical (Dyson swarms, terraforming gas giants, star-lifting). The Kardashev Type I threshold ($\sim 10^{16}$ W) is within reach. Civilisational risk: self-replicating swarms gone wrong; societal dependency on AI decision-making; existential fragility of a highly interconnected system.

Superscience possibilities: Matter transmission without receiver, regeneration rays, reality disruptors, stasis fields, time flow alteration.

TL13+: Devices resemble TL12 but half the mass and cost. Science at this level catalogues what exists rather than explaining it.

3. Technology Paths (Campaign Genres)

These are design templates, not prescriptions. Most real settings blend elements.

Path	TL	Notes
Conservative Hard SF	9 only	No new gadgets after TL9; slow or medium progression
Radical Hard SF	10–12	Most or all superscience omitted; medium/fast/accelerated pace

Path	TL	Notes
Cyberpunk	8–10	Medical/biotech/cybernetics 1–2 TLs ahead of general; neural interfaces and volitional AI common
High Biotech	8+	Biochemicals, biotech, medical TL ahead; inorganic chemistry behind
Nanotech Revolution	9+	Mature molecular nanotech arrives at TL9 instead of TL11; produces singularity risk
Emergent Superscience	Any	Single major superscience breakthrough (e.g., FTL at TL9) reshapes everything else
Retrotech	6–7 base + superscience	Stardrives, blasters, and giant robots grafted onto a lower base
Safe-Tech	10–12 hardware	Cybernetics/AI/nanotechnology suppressed or banned; slow/retarded progression
Psi-Tech	Any	Psychotronics integrated into society; slow/retarded progression common

4. Scale, Transport, and Political Units

Transportation speed defines the practical size of political units (from *GURPS Future History*):

TL	Transport speed	Political unit radius
0	3 km/day walk	Village (~3 km)
1	~8 km/day	Nation (~5,000 km)
5	Rail/steam	Large nation (~27,500 km)

TL	Transport speed	Political unit radius
9	Chemical rockets	~375 km/s (2.2 AU political reach)
10–12	Fast interplanetary	Planet + moons as single political unit
^ (superscience FTL)	Faster-than-light	Multi-system polities possible

Astronomical engineering by TL:

- TL9: interplanetary travel, orbital colonies, asteroid mining (near-Earth), geoengineering
- TL10: fast interplanetary travel, asteroid habitats, simple terraforming (Mars-class)
- TL11: interstellar travel (slow/generation), Venus-class terraforming (difficult)
- TL12: shifting planetary orbits, Dyson swarm construction, planetary disassembly, star-lifting
- TL^ (superscience): antigrav, reactionless drives, FTL, solid Dyson sphere

5. Energy and Population

Energy per person (multiples of human metabolic rate ~100 W):

Era	TL	Multiple
Paleolithic	0	~2×
Neolithic	0	~4×
Ancient empires	1–2	~8×
Medieval	3–4	~16×

Era	TL	Multiple
Industrial Revolution	5	~32×
Electrical/petroleum	6	~64×
Nuclear/digital	7–8	~128×
TL12 projection	12	~256×

Population doubling times:

Era	TL	Doubling time	Rate/decade
Paleolithic	0	2,500 years	0.3%
Agrarian	1–4	1,000 years	0.7%
Industrial/digital	5–8	50 years	15%
Space expansion	9–10	1,000 years	0.7%
Mature nanotech	11	200 years	3.5%
TL12	12	100 years	7.2%

The slowdown at TL9–10 is not decline but investment: habitat-building, terraforming prep, and life extension absorb capital that would otherwise drive population growth. The rise at TL11–12 reflects life extension and reduced mortality overwhelming reduced birth rates.

Earth's carrying capacity at TL5 average is ~625 million. Current (TL8) population is approximately 3× that limit, sustainable only through industrial food systems and fossil energy imports.

6. Future History Dynamics

Forces of Change

Population pressure (Malthus → migration → cultural divergence): New habitable territory permits expansion with characteristic cultural markers. Migrant languages and genetics diversify. When no new territory is available, intensive agriculture intensifies, food imports grow, and city populations concentrate — all increasing the demand for specialised labour and trade.

Economic activity (Kondratiev waves): 50–60 year cycles of expansive investment followed by contraction and creative destruction. New TLs tend to be reached during expansive phases. The four economic phases (Awakening → Unraveling → Crisis → High) also shape social mood and political tolerance for risk.

Generational cycles (Strauss-Howe): ~80-year four-generation cycle (Crisis, High, Awakening, Unraveling) mapped onto cohort psychology. Useful for predicting social mood without deterministic plot constraints.

Civilisational cycles (Spenglerian, ~1,500 years): Spring (heroic/founding) → Summer (aristocratic courts) → Autumn (urbanisation, popular government, Enlightenment) → Winter (bureaucratic mass society, decay). Each phase has characteristic political forms and heroic archetypes.

Illuminated cycles (5 phases): Chaos (anarchy CR0–1) → Discord (class struggle CR3–5) → Confusion (revolt, democracy CR2–4) → Bureaucracy (corporate/technocratic state CR4–6) → Aftermath (skepticism, collapse toward anarchy CR1–5). Then repeats.

Natural Cycles

Climate cycles: 100,000-year glacial/interglacial cycles modulated by Milankovitch orbital mechanics. Current interglacial began ~12,000 years ago. Greenhouse gas emissions may delay or prevent the next glaciation. Multi-century warm/cold phases (Roman Warm Period, Medieval Warm Period, Little Ice Age) affect agriculture and political stability.

Supercontinent cycle: 300–500 million years. Aggregation = low sea level, dry continental interiors, mass extinctions. Dispersal = higher sea level,

warmer seawater, higher biodiversity. Plausible for alien world design.

Stellar evolution: The Sun becomes a red giant in ~5 billion years, making Earth uninhabitable ~1 billion years before that (wet greenhouse from luminosity increase). C4 plants replace C3 ~600 million years from now as CO₂ falls. All life ends without geo-engineering in ~1.2 billion years.

Extreme Events

Technological disasters scale with TL: Fire (TL1), gunpowder mill explosions (TL3), steam engine boilers (TL5), nuclear accidents (TL7), runaway nano-replication ("grey goo") (TL9).

Astronomical disasters: Asteroid impact, gamma-ray burst, rogue planetary flyby, stellar flare. GM tools: 3d15 = no lasting impact; 16–17 = historical disaster; 18 = massive disaster with global impact. Roll for once-per-10-year events.

Plagues and invasive species: Become interplanetary at TL9+. Microorganisms carried between planets on meteoroids or ships. New selective pressures on alien worlds could drive speciation within decades to centuries.

Violence: Small-scale is background noise; war between major powers can be historically transformative. Military resources formula: $MR = \text{Population} \times \text{Monthly Income} \times \text{Military Budget Factor (MBF)}$. MBF ranges from 0 (CR0, none) to 50% (CR6 wartime). In interstellar war, logistic overhead for interstellar transport (200% or more) makes long-term conquest difficult without local supply chains.

Singularity / mutants / superbeings: Vinge's concept of superhuman intelligence — AI or cybernetically augmented — as the event horizon of predictable future history. Beyond it, transcendent beings might resolve history's conflicts instantaneously, making conventional future history frameworks obsolete.

Innovations (Three Phases)

1. **Concept:** The idea that a thing *can* be done in a specific way. May require prerequisite knowledge in unrelated fields. Cannot be rushed by funding alone.
2. **Prototype:** First working model. May be done by lone inventors (Franklin, Wright) or large teams (Manhattan Project).
3. **Production:** Manufacturing at scale with established retail price. Only production raises a society's actual TL.

Kondratiev wave expansive phases cluster innovations (concept → production in < 10 years); contractive phases force out old industries.

Changes of State

Resolution of conflicts leads to things becoming permanently true or permanently false. Useful design question: what has disappeared from your setting that used to exist? (Great powers reduced to secondary status; religions vanished; formerly cheap commodities now scarce.) And what has re-emerged? (Bacteriophage therapy; cosmism; pre-industrial craft traditions.)

7. 2300AD as a Concrete Case Study

Setting Parameters

Year: 2300 AD. Post-Twilight War (decades of conventional and nuclear conflict, late 22nd century). Humanity recovered from ~50% industrial capacity loss and rebuilt.

Political structure: France is the dominant power; other major nations include Manchuria, America, Britain, Azania (South Africa), Germany, Canada, Australia. 57 colonies on 32 worlds. 75% of humanity on Earth; 90% on Earth + Tirane (Alpha Centauri) combined.

FTL: Stutterwarp drive, 0.5–9.5 light years per day (typically 1–2 LY/day). Requires ~40 hours near a gravity well to discharge. Maximum range 7.7 LY per jump. The three arms of exploration (French, American, Manchurian) are shaped by this range limit. No reactionless drives, no antigrav.

Three arms: French Arm (French-dominated), American Arm, Manchurian Arm. Most human colonies lie within a few jumps of Earth along these corridors.

Other constraints: No volitional AI (expert systems only); human creativity and intuition are still premium labour. No artificial gravity except spin. Aerodynes use vectored thrust. Average lifespan ~120 years (ageing rolls start at 50). Six known alien species; two star-faring.

TL stratification:

Context	2300AD designation	GURPS TL
Frontier/colonial standard	Old Commercial (OC)	~10
Core worlds, large corporations	New Commercial (NC)	~11
Top militaries, TransNationals	New Military (NM)	~12

This two-TL spread between frontier and cutting-edge is a deliberate design choice representing the lag between metropole innovation and colonial adoption.

Representative Equipment by TL

TL9 (surplus/legacy):

- SG-77 Sturmgewehr (5.5mm conventional assault rifle, Germany)
- FC-68 sporting rifle (5mm, France), Wolf hunting shotgun, MG-7 machinegun
- P-11 pistol; SS-7 air rifle; Stracher Wolf shotgun

TL10 (frontier standard):

- *Personal*: Respirator (filter mask), cold weather clothing, backpack, water purifier, fuel cell (25 kg, 10-hr, Power 1), small fabricator (small parts, 2–3 min print time), basic tool kit, excavating tools, construction tools
- *Comms*: Hand communicator (20 km range), backpack communicator (200 km), tight-beam uplink to orbit
- *Sensors*: Binoculars (thermal imaging, 50× magnification), large life form detector (IR, 100 m increment), remote meteorological station
- *Medical*: Medkit (spray bandages, autoinjector, antishock, antibiotic)
- *Power*: Fuel cell, fuel station (solar electrolysis of water → hydrogen)
- *Industrial*: Light industrial fabricator (100×100×30 cm max object, 10–15 min print), Pichot 4020 HMHPAT exo-loader (15× strength amplification)
- *Weapons (infantry)*: FAM-90 gauss rifle (4.5mm, 1,600 mps, burst-fire DM+1), Sk-19 binary propellant assault rifle (Germany), Type-49 assault rifle, various conventional rifles and handguns, Mini-12 heavy machinegun; Plastique-9 explosive
- *Civilian weapons*: FC-70 hunting rifle, Model-10 riot gun, standard pistols (9-23, 12-39)

TL11 (core world / advanced commercial):

- *Comms/computing*: Portacomp (computer/4, 0.5 kg, Lv300) — handheld programmable, wearable, neural jack-capable; Link Phone (computer/1, smartphone equivalent with GPS and Link network access); Computer Workstation (computer/5, portable but large)
- *Tools*: Electronic repair tools; advanced locksmith kits; Tous-Outil multi-tool
- *Special equipment*: Stik-kit (van der Waals adhesive patches); spinner (carbon monofilament, 0.1 mm, 1,000 kg support)
- *Sensors*: Basecamp security sensor (500 m ground / 5 km aerial detection), SCEMS electromagnetic sensor (1 km range)

- *Medical*: Assistive exoskeleton (STR +2 to +6); lightweight autodoc (4× healing rate, 45 kg, Lv6000); MetabolX drugs (metabolism ×0.1 to ×0.001 for hibernation/healing); Vasopressin-Y (INT boost, addictive); Interferon-Gamma (anti-viral, stops DNAM retrovirus within 24 hours)
- *Weapons*: SVB laser rifle (Russia, 500 m, 5D, Scope), Type 1 plasma gun (Manchuria, 100 m, 6D, AP 10); AS-89 gauss rifle (Russia); Modell-6 pistol (AP 4); Luce-7 laser rifle; Stracher MP-67/MP-67K PDWs; Traylor T-50
- *Colonisation (Soft Path)*: DNA modifications (DNAMs) for planetary adaptation; 0G DNAM for spacers; "More Hands" foot variant

TL12 (top military / TransNational):

- *Weapons*: FTE-10 sniper (10mm gauss, 1,200 m, 6D, AP 10); FAM-90 (French standard gauss rifle); Sk-19 (binary propellant, Germany); F-44 gauss machinegun (4.5mm, 800 m, 4D+1, Auto 6); Mueller-Rivera F-19 laser rifle (400 m, 6D, Scope); Lk-1 laser rifle (Germany); P-3 laser pistol (Auto 3); Mk 2-A2 PGHP plasma gun (20MW, 350 m, 10D, AP 18); A-9 plasma Sturmgewehr (50 m, 4D, AP 8); M-97 binary machinegun (AP 5, Auto 4); wire knife (hyperdiamond monofilament loop)
- *Medical*: Static autodoc (1,000 kg, 6× healing rate, DM+2 to Medic, autonomous operation); Radox (acute radiation exposure mitigation)
- *Surveillance*: Thermal-visual camouflage (colour-matching electronic paper + heat redistribution, DM+6 Stealth stationary)
- *Pentapod biological tech*: Biocontacts (IR night vision, 5× magnification, live organism worn in eye — TL16 by Pentapod standards); Water Breather (aquatic organism worn as rebreather); Biosampler (detects food safety)

Ghost Diamonds (late 2280s): Deceased persons' digital "footprint" used to construct diamond-film memory with simulated personality capable of passing the Turing test. Not a mind upload — a simulation built from data artefacts. A philosophical/legal flashpoint.

Colonisation Philosophy

Hard Path: Technological adaptation. Machinery, filter masks (respirator, TL10), greenhouses, carniculture (vat-grown meat), pressure tents (TL10–11). The colony brings Earth's environment with it.

Soft Path: Genetic adaptation. DNAMs (DNA Modifications) tailored to specific worlds. Requires biotech infrastructure and commits colonists biologically to a world. Creates sub-species over generations. In GURPS terms: planetary adaptation DNAM packages bought at character creation.

Spacers: 0G DNAM — adapted to microgravity. "More Hands" — prehensile foot for micro-gravity manipulation. Legally and culturally distinct from planetary-gravity humans on many worlds.

8. Transhuman Space — The TL9/10 Transition

Source: GURPS Transhuman Space Classic. Setting year: 2100 (late TL9, early TL10). The TS framework is the most detailed treatment of the biological, digital, and social consequences of the first century of transhuman technology.

The Fifth Wave

Human civilisation has undergone four historical waves: agricultural (Neolithic), industrial, information, and biotech/nanotechnology. The emerging Fifth Wave represents the convergence of mature nanotechnology and sapient AI — the point where human creativity is no longer the binding constraint on production. The Fifth Wave is not inevitable at any particular TL; it requires social and political conditions, not merely technical ones.

AI Taxonomy

TS Space defines a three-tier AI hierarchy that avoids the binary "dumb tool / dangerous superintelligence" framing:

Designation	Description	Legal status (2100)
NAI (Non-sapient AI)	Task-specific, no self-awareness; follows instructions without initiative	Property everywhere
LAI (Low-sapient AI)	Self-aware but limited; capable of learning, preference, and limited initiative	Property in most jurisdictions
SAI (Sapient AI)	Full self-awareness, creativity, emotional range; indistinguishable from biological intelligence in conversation	Rights in EU and Islamic Caliphate; property elsewhere

By 2100 approximately 100 million SAI exist. The legal fault line between SAI-as-property and SAI-as-person is the most significant civilisational conflict in the setting.

The Receding Singularity: Moore's Law slowed in the 2020s. Human biological augmentation accelerated at approximately the same rate as AI development, preventing the projected AI capability explosion. There is no singularity by 2100 — only a slow escalating competition between human (and posthuman) intelligence and machine intelligence. Both improve; neither dominates.

Mind Emulations

The ability to run a human mind as software is TL10 in the standard GURPS scale; TS Space treats it as borderline TL9/10, achievable by 2100 with significant constraints.

Ghost (full emulation): Complete mind transfer via destructive uploading. Process: nanostasis (halt biological processes) + bush robot disassembly + Ghost Compiler (10 hours). The original brain is destroyed; the result carries full DX, IQ, mental advantages, skills, and memory. A Ghost is legally distinct from the biological original and has no right of inheritance in most jurisdictions.

Fragment: Incomplete upload — halved mental skills, partial amnesia. Caused by suboptimal nanostasis or incomplete compilation.

Shadow: Non-destructive emulation derived from observation rather than direct substrate copying. Lower fidelity; may have personality drift from the original.

Xoxing: Creating multiple instances of the same Ghost (copying). A Ghost and its xox are legally treated as distinct individuals with no special relationship.

Infomorphs and Shells

An **infomorph** is a digital intelligence without a fixed substrate — it exists as software and can migrate, copy, and instantiate on any sufficient computing hardware. The body becomes optional.

Cybershell: Mechanical body inhabited by an infomorph. Ranges from humanoid to industrial to microscale microbot swarms.

Bioshell: A vat-grown biological body inhabited by a Ghost or SAI. Allows infomorphs to interact with biological-social structures without the friction of obviously mechanical bodies.

Parahumans and Pantropy

Parahuman: A human whose genome has been transgenically modified with non-human genes, creating stable biological variations — faster reflexes, enhanced smell, radiation resistance, cold tolerance. Parahumans breed true and are legally human in most jurisdictions.

Pantropy: Adapting life to fit environments rather than adapting environments to fit life. The inverse of terraforming. On a low-gravity world, instead of building habitats maintaining Earth-standard gravity, you engineer humans to live comfortably at 0.2g. Pantropy is cheaper than terraforming at TL9–10 and becomes the pragmatic choice for early colonisation.

The calcium hack: Zero-G adapted parahumans require a metabolic modification that decouples skeletal density maintenance from gravitational stimulus. This DNAM makes long-duration microgravity habitation economically viable — spacers do not lose bone and muscle mass in freefall, eliminating the medical overhead that makes current microgravity habitation expensive.

Bioroids and Sapient Uplifts

Bioroid: A vat-grown biological construct with partial human genome, designed for specific roles (military, heavy labour, hazardous environments). Legally property; not legally human even if sapient.

Sapient Uplift: Engineering animals to full sapience via genetic modification. Primary candidates: cetaceans, great apes, octopuses, ravens. An uplift species is legally sapient if it demonstrably passes sapience tests.

Memetics

Memetics (the science of how beliefs propagate) becomes an engineering discipline at TL9.

- **Memetic engineering:** Designing belief complexes that spread virally within target populations. Used by corporations, governments, religions, and political movements.
- **Memetic warfare:** Deploying belief-system attacks against enemy populations to erode morale, change voting behaviour, or induce mass defection.
- **Countermemetics:** Defensive systems and educational frameworks to resist memetic attack.

Ideological Fault Lines

Nanosocialism: The political argument that all patents and information should be publicly owned, since information goods are non-rival and monopoly patents on basic nanotechnology concentrate power unjustly. A

significant political force in 2100; the EU's partial implementation creates tension with corporate-IP regimes elsewhere.

Pan-sapient rights: The political movement extending legal personhood to all sapient entities — SAI, uplifted animals, Ghosts, bioroids. Opposed by economic interests that depend on property-status labour. The EU and Islamic Caliphate are ahead of other polities in granting rights; China and the US treat most non-biological sapients as property.

Duncanites: Anarchist/libertarian inhabitants of the outer belt who deliberately push technological limits outside regulatory oversight. Early adopters of illegal modifications, experimental memetics, and unapproved AI architectures. They are both a safety hazard and an innovation engine — prohibited technologies developed in the belt often become mainstream within a generation.

9. FTL Spectrum and Political Implications

FTL speed directly determines the scale of viable political units. The spectrum from slow to fast fundamentally changes what interstellar civilisation looks like.

The Spectrum

FTL regime	Speed	Political analogy	Communication lag (50 LY radius)
Slow (this projection's baseline)	6 months/LY = 2 LY/year	Ancient maritime empire	25 years each way
Medium (2300AD stutterwarp)	0.5–2 LY/day	19th-century oceanic empire	~50–100 days each way

FTL regime	Speed	Political analogy	Communication lag (50 LY radius)
Fast (this projection's endpoint)	1 day/LY = 365 LY/year	Colonial-era steamship empire	~50 days each way
Very fast (jump/hyperdrive)	1–7 LY/hour	Modern telecommunications empire	Hours to days

Slow FTL (6 months per light year)

Alpha Centauri (4.2 LY) requires ~2 years transit each way — a 4-year round trip for a message and reply. A 50-LY empire reaches hundreds of worlds but its edge is 25 years distant. This is roughly the political situation of the Roman Empire relative to its frontiers, stretched across three spatial dimensions.

Political consequences:

- No interstellar empire is viable by military force; colonies are functionally independent within 1–2 generations
- Trade is high-value-only (information, unique biological samples, luxury goods, expertise in transit)
- Cultural divergence is guaranteed within 3–5 generations per colony
- The most powerful interstellar "empire" is effectively a cultural tradition and trading network
- Military response time to colonial rebellion is measured in decades

Economic consequences:

- Self-sufficiency is a survival requirement, not a political preference
- Comparative advantage is extreme — a world with a unique resource charges monopoly prices for decades before competitors can establish

themselves

Medium FTL (2300AD stutterwarp, ~1–2 LY/day)

The 40-hour gravity-well discharge requirement gives an effective speed of ~0.3–1 LY/day for complete journeys. Alpha Centauri is ~8 days; a 50-LY radius is 100 days at best. Analogous to 19th-century oceanic travel: colonies develop genuine autonomy and cultural identity, but meaningful trade and delayed imperial oversight are both possible. This is the 2300AD structure: France can exercise real influence over its colonies, but colonial administration must be substantially autonomous.

Fast FTL (1 day per light year)

Alpha Centauri: ~4 days. A 50-LY empire: ~50 days maximum transit. A 500-LY empire: ~500 days.

Genuine interstellar administration becomes viable: fleets can reach rebellions before they consolidate. Empires and federations with real enforcement power are possible. This is approximately the Star Wars / Star Trek political scale.

The FTL transition as a historical event: The shift from slow to fast FTL is comparable to the shift from sail to steam/rail in the 19th century. Political consequences are dramatic: previously autonomous colonial worlds suddenly find themselves within reach of the metropole's military. This transition is a natural narrative break in a 10,000-year future history — the moment when isolation ends and integration (by trade or by force) becomes unavoidable.

Infomorph Transmission vs. Physical FTL

A frequently overlooked asymmetry: infomorphs (uploaded minds) can travel as data. Any interstellar communications link — laser, radio, quantum relay — transmits an infomorph at light speed, bypassing FTL travel entirely.

- At slow FTL (2 LY/year), an infomorph transmitted by tight-beam laser travels at the speed of light (1 year/LY) — only 2× faster than a ship, but with zero physical cargo overhead
 - At slow FTL, the question "should I send a person or an infomorph?" is economically non-trivial
 - Infomorph-heavy civilisations can maintain faster information networks than physical-travel civilisations at any FTL speed below lightspeed
 - The information economy of a slow-FTL star empire may run largely on infomorph/data transmission while biological populations and physical goods move much more slowly
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10. Speculative Resources on the Long Run

A working bibliography for 10,000-year future history design. Organised by conceptual contribution.

Civilisational Scale and Collapse

Isaac Asimov — Foundation series (1942–1993): The defining treatment of empire-scale future history. Psychohistory — statistical prediction of mass human behaviour — is the device that makes a 30,000-year arc tractable. Core tension: the Galactic Empire collapses despite Hari Seldon's calculations; the question is whether the Dark Age is 30,000 or 1,000 years. Key concepts: momentum of civilisational collapse; the gap between prediction and intervention; hidden information-state actors as civilisational backstops.

Olaf Stapledon — Last and First Men (1930): Canonical treatment of deep-time human evolution. Eighteen distinct species of humanity over 2 billion years, each arising from the ashes of the previous civilisation. No FTL; the relevant lesson is that at sufficient timescale, evolutionary and

social selection creates beings qualitatively unlike the original — not divergent versions of us, but genuinely new kinds of minds.

Stapledon — Star Maker (1937): Extends to cosmic timescales.

Civilisations arise, peak, fail, and are replaced. Design concept: civilisation as experiment; the universe as indifferent to any particular civilisation's survival.

Post-Human and AI Transition

Vernor Vinge — "The Coming Technological Singularity" (1993) and *A Fire Upon the Deep* (1992): The Singularity as event horizon of predictable history. In *A Fire Upon the Deep*, different regions of the galaxy have different physics that constrain intelligence levels — the Zones of Thought. Near the galactic core (Unthinking Depths), FTL and superintelligence are impossible; at the galactic rim (the Beyond), they are trivial. A structural device for incorporating the Singularity into a playable setting without having it terminate the story.

Iain M. Banks — Culture series (1987–2012): The post-scarcity post-human endpoint. The Culture is run by benevolent superintelligent AIs (Minds); biological humans are passengers who matter morally but not economically. Key concept: what happens to biological humans when they are no longer economically necessary? Banks's answer — hedonism, games, and occasional heroism as ambassadors to less advanced civilisations — is one plausible endpoint for Path 3 (post-human).

Charles Stross — *Accelerando* (2005): Near-future acceleration to singularity traced through three generations of one family. The solar system is consumed and converted into computronium. Most biological humans flee to the outer system. Key concept: the singularity doesn't kill everyone — it relocates them economically.

Divergence and Speciation

Gordon Dickson — Childe Cycle (Dorsai!, 1960–1988): Thirteen colonised worlds, each founded by a specific cultural/philosophical fragment of humanity. Each colony develops the founding characteristic to an extreme: the Dorsai are perfect warriors; the Exotics are philosophical/psionic. Key concept: colony worlds as controlled experiments in cultural selection pressure; humanity splinter-culturing its way toward civilisational plurality.

Kim Stanley Robinson — Mars Trilogy (1992–1996): Ground-level treatment of terraforming (centuries of real work), institutional decay, life extension politics, and biological ethics. Key concept: the political economy of very long human lives in a resource-limited frontier environment; colonists diverge socially within two generations, not biologically.

Alastair Reynolds — Revelation Space series (2000–2007): Hard SF with no FTL. The Fermi paradox is resolved by the Inhibitors — machines designed to suppress any civilisation reaching a critical threshold of space-industrialisation. Human civilisation fragments into Demarchists (democratic technocrats), Conjoiners (neural-networked hivemind), and Ultras (ship-dwelling transhumans). Key concept: divergence into incompatible post-human factions as the natural result of different technological paths; the Inhibitor hypothesis as a structural solution to the silence of the stars.

Political Economy at Stellar Scale

Peter F. Hamilton — Commonwealth Saga (2004–2005): ~300-year near-future with rejuvenation (humans routinely live 200+ years), wormhole train network connecting 600 colony worlds, and corporate/nation-state politics at interstellar scale. Key concept: rejuvenation politics — people who live 300 years do not relinquish power; the gerontocracy problem.

Cordwainer Smith — Instrumentality of Mankind stories (1950–1966): Deep-future setting spanning ~15,000 years. The Instrumentality are near-immortal rulers of a stagnant but stable human civilisation. Underpeople are uplifted animals with human intelligence but sub-human legal status. Key

concept: the natural endpoint of a stable stratified civilisation sustained by biological hierarchy; Spenglerian Winter given a face.

Paul J. McAuley — The Quiet War series (2008–2012): Near-future colonisation of the outer solar system by radical biopunks who practice pantropy on themselves and their descendants. Key concept: how quickly pantropy creates genuinely alien beings from human stock — faster than people expect.

Non-Fiction and Theory

Nick Bostrom — Superintelligence (2014): Rigorous treatment of the AI control problem. Orthogonality thesis: intelligence does not imply values compatible with human welfare. A sufficiently capable AI pursuing any goal will resist modification and acquire resources without limit.

Ray Kurzweil — The Singularity Is Near (2005): Accelerationist position. Exponential growth in computing continues; biological/digital merger is inevitable by ~2045. The Singularity as a tractable engineering project rather than an unknowable event horizon (opposed to Vinge's formulation).

11. The 10,000-Year Projection

Parameters: slow FTL begins at 6 months/LY (2 LY/year); fast FTL (1 day/LY) becomes available mid-projection. Setting is a human star empire with hundreds of worlds. Three human paths run in parallel and in tension throughout.

Three paths:

- 1. Persistent humans — Homo sapiens baseline; technology is a tool; biological form conserved*
- 2. Divergent humans — pantropy and genetic selection; biological adaptation creates sub-species*
- 3. Post-humans — AI, uploading, and transhuman transformation*

Era 1: Foundation (Years 1–2,000)

Physical context: Slow FTL (2 LY/year). Alpha Centauri takes ~2 years. A 50-LY empire is a 25-year journey end-to-end. Political communication lags are decades. TL at Era start: TL10 (robotic age). By year 2,000, TL11 is available at core worlds.

Path 1 — Persistent: Safe-Tech laws suppress the most disruptive AI and nanotechnology. Hard Path colonisation dominates: pressure suits, fabricators, filtered environments, carniculture. Colonies diverge *culturally* rapidly — within 3–5 generations, colonial worlds have distinct languages, religions, political forms — but not *biologically*. Trade is luxury-goods only; political bonds are attenuated. By year 2,000, most Persistent-path worlds are effectively independent nation-states with ceremonial allegiance to a distant metropole. The political structure resembles a loose Hanseatic League with a shared trading language and common origin mythology.

Path 2 — Divergent: Soft Path colonisation via DNAM packages begins immediately. Worlds with radically non-terrestrial conditions require aggressive genetic engineering within the first generation to survive without expensive life-support infrastructure. By year 500, distinct biological variants exist: vacuum-adapted spacers (calcium hack + radiation tolerance), high-G variants (denser musculature, shorter stature, reinforced skeletal structure), aquatic-world variants (gills, altered thermoregulation). By year 2,000, some variants have been reproductively isolated for 40+ generations; speciation is not complete but interbreeding with baseline produces low fertility. Divergent-path worlds regard baseline humans as a related species — close enough for cultural exchange, distant enough for legal ambiguity about citizenship and rights.

Path 3 — Post-human: The TS Space Receding Singularity dynamic plays out. Ghost technology enables digital immortality and xoxing. The first worlds with legal SAI personhood allow infomorphs full citizenship. By year

500, some worlds are primarily infomorph populations. By year 1,000, the first "substrate worlds" appear — asteroid habitats converted almost entirely to computing substrate, inhabited by billions of infomorphs in fractions of a second subjective time per second objective. By year 2,000, post-human worlds have a TL12 understanding of reality that makes them approximately incomprehensible to Persistent-path TL11 worlds. Cultural contact exists but requires deliberate translation effort.

Era 1 tensions:

- Metropole attempts to reassert control via expensive military expeditions (25-year response time makes this ineffective and ruinous)
 - Biological-variant colonies seek legal recognition that Persistent-path worlds resist
 - Post-human worlds are technologically dominant but geographically marginalised (asteroid belts and non-habitable systems, not prime agricultural planets)
 - Infomorph transmission networks (light-speed) outrun physical FTL ships in information delivery, giving post-human worlds a communications advantage
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Era 2: Integration (Years 2,000–6,000)

Physical context: Fast FTL (1 day/LY) becomes available at approximately years 2,000–2,500. The technology requires TL11+ industrial infrastructure to manufacture, and metropole controls access. This creates a decade-scale window of political instability as the first fast-FTL powers project military force onto worlds that previously had 25 years of warning before a fleet arrived. TL12 is achieved at core Post-human worlds by approximately year 2,500.

Path 1 — Persistent: Fast FTL triggers a re-integration crisis. The metropole attempts to reassert sovereignty over colonies that have been de

facto independent for 40+ generations, producing integration wars (analogous to 19th-century colonial reconquests). A new political equilibrium emerges by year 3,000: a loose interstellar federation of Persistent-path worlds with a shared legal framework, regular interstellar congresses, and genuinely enforceable trade agreements — but still no government capable of overriding local sovereignty on internal matters. The federation persists because fast FTL makes collective security viable. By year 6,000, Persistent humans still look like *Homo sapiens*; lifespan is 200–400 years via anti-ageing technology but the body remains biological.

Path 2 — Divergent: The fast-FTL transition primarily affects Persistent-path politics; Divergent-path worlds are largely peripheral to the reconquest wars because they occupy marginal worlds the Persistent polities don't consider worth the military effort. By year 3,000, many Divergent-path populations qualify as distinct species: reproductive isolation is complete, cognitive style is different (high-G variants process spatial information differently; zero-G variants have distinct vestibular systems). By year 6,000 there are 15–30 recognisably distinct Divergent human species, each world-adapted, each with 100+ generations of cultural evolution. The Divergent-path worlds form a loose ecological network — trading biological techniques, DNAMs, and survival expertise — rather than a political alliance.

Path 3 — Post-human: Post-human civilisation at year 2,000 has already achieved TL12 in substrates optimised for infomorph existence; by year 3,000, leading substrate-worlds are approaching TL13. Physical contact with post-human worlds is still possible but has the character of visiting a museum: post-humans maintain interfaces for biological visitors as a courtesy, not a necessity. By year 6,000, post-human civilisation is no longer comprehensible as a variant of human civilisation.

Era 2 tensions:

- Persistent-path integration wars establish the precedent for fast-FTL military power overwhelming slow-FTL worlds

- Divergent species claim political sovereignty that Persistent federations are reluctant to grant (they look human-ish; they are treated as colonial minorities rather than independent peoples)
 - Post-human worlds are technologically superior to everyone but have no interest in physical conquest
 - Biological humans who upload and emigrate to post-human worlds are legally dead in most Persistent-path jurisdictions, creating a brain-drain/emigration crisis
 - "Infomorph missionaries" from post-human worlds offer uploading to Persistent populations; some worlds ban this; others permit it; a few make it mandatory
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Era 3: Fracture or Transcendence (Years 6,000–10,000)

Physical context: TL12 is universal at leading edges. Physical substrate (planets, asteroid belts) remains relevant for Paths 1 and 2; Path 3 may have moved toward Dyson swarms or stellar engineering projects.

Path 1 — Persistent: After 10,000 years the Persistent-path federation(s) face Spenglerian Winter. Institutions revolutionary in year 1,000 are calcified bureaucracies by year 8,000. Lifespan extension means political leaders entrench for centuries; the gerontocracy problem becomes existential. Two scenarios:

- *Scenario A — Enduring Civilisation:* A periodic institutional reset mechanism (forced sabbatical, term limits backed by force, revolutionary generations every ~800 years following Spenglerian cycles) prevents total calcification. Persistent humans at year 10,000 are stable, conservative, TL12, and somewhat alien to their TL8 ancestors but recognisably continuous. Lifespan 1,000–2,000 years.
- *Scenario B — Fracture:* The federation fragments under accumulated institutional debt, gerontocratic entrenchment, and the sheer cultural

diversity of 10,000 years of colonial evolution. Year 10,000 Persistent-path humanity is a mosaic of hundreds of world-state polities with a dead common language (like Latin in medieval Europe). Individual worlds may sit at TL10–14 depending on local history.

Path 2 — Divergent: At year 10,000, humanity has speciated into perhaps 50–100 distinct biological forms, each adapted to its world. These are not variants — they are species, reproductively isolated, cognitively distinct, culturally independent for 200+ generations. Some are post-sapient (TL13+ cognitive enhancement creating minds qualitatively beyond baseline human); some are conservatively natural (rejecting all genetic engineering after the original DNAM packages). Collectively, the Divergent path has created genuine civilisational plurality: not cultural diversity within a shared framework, but independent civilisations that share only an evolutionary origin.

Path 3 — Post-human: The post-human trajectory after year 6,000 is genuinely unpredictable from the outside. The most conservative projection is that post-human civilisation has converted most accessible stellar material in inhabited systems to computing substrate, with population in the billions of biological-equivalent minds and trillions of faster-than-biological infomorphs. Three sub-scenarios for the endpoint:

- *Transcendence:* Post-human civilisation departs the observable stellar neighbourhood, pursuing an agenda incomprehensible to biological minds. The Fermi paradox may be answered here: mature post-human civilisations simply leave the accessible frame of reference.
- *Stewardship:* Post-human civilisation remains as a benevolent (or not) background presence for Paths 1 and 2, intervening occasionally and incomprehensibly. The Banks Culture model.
- *Internal Fracture:* Post-human civilisation itself fractures into multiple incompatible successor-minds pursuing incompatible goals. The stellar neighbourhood becomes host to multiple godlike entities whose

interaction with biological humans ranges from indifference to catastrophe. The Reynolds Inhibitor model without the malevolence.

Convergence and Interaction

The three paths do not exist in isolation. Their intersections generate the most historically significant events:

- **The Upload Crisis** (~years 1,000–1,500): Biological humans on Persistent and Divergent worlds choose uploading at significant rates. The legal and demographic consequences are permanent.
 - **The Speciation Debates** (~years 2,500–3,500): When do Divergent humans stop being human? The legal answer (reproductive isolation + 50+ cognitive differences) satisfies biologists and infuriates diplomats.
 - **The Silence of the Post-humans** (~years 6,000–7,000): Post-human worlds stop responding to conventional communication. Are they dead, transcended, or simply uninterested? The uncertainty drives Persistent and Divergent civilisations toward consolidation or panic.
 - **The Return** (~years 8,000–9,000): One or more post-human entities re-contact biological civilisations with an offer, a warning, or a demand. The content determines whether this is the greatest gift or greatest catastrophe in biological-human history.
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12. Synthesis Notes for Building a Future History

Choosing a TL

Question 1: What is the *bottleneck* technology? A setting at TL11 overall but with suppressed AI (Safe-Tech variant) looks very different from a standard TL11. The bottleneck shapes social structure.

Question 2: What is the *lag*? Core → frontier technology lag in 2300AD is 1–2 TLs, which creates built-in inequality and adventure opportunity. A more

egalitarian setting might have < 0.5 TL spread; a Balkanised one might have 3+ TL spread between factions.

Question 3: What superscience, if any, exists? A single superscience breakthrough (FTL at TL9, as in 2300AD) reshapes *everything downstream* — colonial settlement, power structure, economics — without requiring high TL across the board.

Anchoring a Date

Use the TL progression table as a baseline, then apply modifiers:

- Post-war recovery: add 50–200 years (2300AD lost ~100 years to the Twilight War)
- Accelerated path: subtract 50–100 years
- Safe-Tech (AI/nano suppressed): add 200–500 years to affected domains

Population and Distribution

At TL10 (medium pace, ~2120), Earth population may have peaked and begun slow decline (post-industrial demographic transition) while off-world population grows. 2300AD: 75% of ~20 billion on Earth, 25% spread across 32 worlds — an early, thin diaspora. At TL11 the ratio might invert over centuries as life extension and off-world birth rates diverge.

Social Structure

Transportation determines political unit size. Stutterwarp (1–2 LY/day with discharge overhead) makes interstellar travel roughly analogous to 19th-century oceanic travel: possible but expensive and slow enough that colonies develop genuine autonomy. This is the design kernel for 2300AD's nation-state multipolar structure rather than a single interstellar empire.

At genuine FTL (days to anywhere), interstellar empires become viable. Without it, any multi-system polity requires either enormous patience or a

different enforcement mechanism (economic rather than military).

Cycles and Events

When placing events in a future history timeline, layer:

1. **Kondratiev-scale** economic waves (50–60 years): innovation clusters and busts
2. **Generational cycles** (~80 years): social mood shifts, ideological reversals
3. **Spenglerian cycles** (~1,500 years): civilisational rise/autumn/winter — useful for deep-future settings
4. **Extreme events** (roll 1/decade): plagues, asteroids, technological disasters, wars, alien contact

2300AD uses a single extreme event — the Twilight War — as the defining break that reset the political map. The war was severe enough to kill off the USA and USSR as dominant powers and enable France's hegemony. Extreme events create the discontinuities that make futures look genuinely alien rather than merely extrapolated.

The Human Future Question

Three broad paths:

1. **Persistent humans** (Safe-Tech / cultural conservatism): Homo sapiens persists essentially unchanged; technology is a tool, not a transformation. 2300AD occupies this niche.
2. **Divergent humans** (Soft Path expanded): genetic and symbiont adaptation creates distinct sub-species adapted to different environments. Intermarriage and cultural exchange possible but fraught. Dickson's *Childe Cycle* Splinter Cultures; 2300AD Spacers/DNAMs are early examples.
3. **Post-humans** (Singularity / Transhumanist): the transformation of humanity into something functionally incomprehensible to TL8

observers. Future history ends here in the conventional sense; what follows requires different narrative frameworks.

Most durable SF settings occupy position 1 or 2. Position 3 tends to terminate or radically constrain the kind of character-level adventure that makes a game setting playable.